

Docker, Compose & Kubernetes Cheat Sheet

A practical command reference split by work task.

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Docker

1. Images

Command	Description
<code>docker images</code>	List local images
<code>docker pull <image>:<tag></code>	Download an image
<code>docker image inspect <image></code>	Show image metadata
<code>docker image history <image></code>	Show layer history

Command	Description
<code>docker tag <image> <new>:<tag></code>	Tag an image
<code>docker rmi <image></code>	Remove an image
<code>docker image prune</code>	Remove dangling images
<code>docker image prune -a</code>	Remove all unused images

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2. Containers – Run & Lifecycle

Command	Description
<code>docker run <image></code>	Create & start a container
<code>docker run -d <image></code>	Run detached (background)
<code>docker run -it <image> sh</code>	Run interactive with a shell
<code>docker run --name <name> <image></code>	Assign a name
<code>docker run -p 8080:80 <image></code>	Map host:container port
<code>docker run -e KEY=val <image></code>	Set environment variable
<code>docker run --rm <image></code>	Auto-remove on exit
<code>docker run -v <vol>:/path <image></code>	Mount a volume
<code>docker ps</code>	List running containers
<code>docker ps -a</code>	List all containers
<code>docker start <container></code>	Start a stopped container
<code>docker stop <container></code>	Gracefully stop
<code>docker restart <container></code>	Restart
<code>docker kill <container></code>	Force stop
<code>docker rm <container></code>	Remove a container
<code>docker rm -f <container></code>	Force remove (running)
<code>docker rename <old> <new></code>	Rename a container

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3. Inspecting & Debugging

Command	Description
<code>docker logs <container></code>	View logs

Command	Description
<code>docker logs -f <container></code>	Follow logs live
<code>docker logs --tail 100 <container></code>	Last 100 log lines
<code>docker exec -it <container> sh</code>	Open a shell inside
<code>docker exec <container> <cmd></code>	Run a command inside
<code>docker inspect <container></code>	Full JSON details
<code>docker stats</code>	Live resource usage
<code>docker top <container></code>	Running processes
<code>docker diff <container></code>	Filesystem changes
<code>docker port <container></code>	Show port mappings
<code>docker cp <container>:/path ./local</code>	Copy out of container
<code>docker cp ./local <container>:/path</code>	Copy into container

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4. Volumes & Data

Command	Description
<code>docker volume ls</code>	List volumes
<code>docker volume create <name></code>	Create a volume
<code>docker volume inspect <name></code>	Inspect a volume
<code>docker volume rm <name></code>	Remove a volume
<code>docker volume prune</code>	Remove unused volumes
<code>docker run -v <name>:/data <image></code>	Named volume mount
<code>docker run -v \$(pwd):/app <image></code>	Bind mount current dir

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5. Networking

Command	Description
<code>docker network ls</code>	List networks
<code>docker network create <name></code>	Create a network
<code>docker network inspect <name></code>	Inspect a network
<code>docker network connect <net> <container></code>	Connect container

Command	Description
<code>docker network disconnect <net> <container></code>	Disconnect container
<code>docker network rm <name></code>	Remove a network
<code>docker network prune</code>	Remove unused networks
<code>docker run --network <name> <image></code>	Run on a network

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6. Building Images & Dockerfile

Command	Description
<code>docker build -t <name>:<tag> .</code>	Build from Dockerfile
<code>docker build -f <file> .</code>	Use a specific Dockerfile
<code>docker build --no-cache -t <name> .</code>	Build ignoring cache
<code>docker build --build-arg KEY=val .</code>	Pass a build argument
<code>docker buildx build --platform linux/amd64,linux/arm64 .</code>	Multi-arch build
<code>docker commit <container> <image></code>	Image from a container

Common Dockerfile instructions: FROM, WORKDIR, COPY, ADD, RUN, ENV, ARG, EXPOSE, CMD, ENTRYPOINT, VOLUME, USER, HEALTHCHECK.

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7. Registry (Push/Pull/Login)

Command	Description
<code>docker login</code>	Log in to Docker Hub
<code>docker login <registry></code>	Log in to a private registry
<code>docker logout</code>	Log out
<code>docker pull <image></code>	Pull an image
<code>docker push <user>/<image>:<tag></code>	Push an image
<code>docker search <term></code>	Search Docker Hub

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8. System & Cleanup

Command	Description
<code>docker system df</code>	Show disk usage

Command	Description
<code>docker system prune</code>	Remove unused data
<code>docker system prune -a --volumes</code>	Aggressive cleanup
<code>docker info</code>	System-wide info
<code>docker version</code>	Client/server version
<code>docker events</code>	Stream real-time events
<code>docker container prune</code>	Remove stopped containers

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Docker Compose

Modern syntax uses `docker compose` (space). Older installs use `docker-compose`.

9. Compose – Lifecycle

Command	Description
<code>docker compose up</code>	Create & start all services
<code>docker compose up -d</code>	Start detached
<code>docker compose up <service></code>	Start one service
<code>docker compose down</code>	Stop & remove containers/networks
<code>docker compose down -v</code>	Also remove volumes
<code>docker compose start</code>	Start existing services
<code>docker compose stop</code>	Stop services
<code>docker compose restart</code>	Restart services
<code>docker compose pause / unpause</code>	Pause/resume services

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10. Compose – Inspect & Debug

Command	Description
<code>docker compose ps</code>	List service containers
<code>docker compose logs</code>	View all logs
<code>docker compose logs -f <service></code>	Follow one service's logs
<code>docker compose exec <service> sh</code>	Shell into a service

Command	Description
<code>docker compose run <service> <cmd></code>	One-off command
<code>docker compose top</code>	Running processes
<code>docker compose config</code>	Validate & view merged config
<code>docker compose events</code>	Stream events

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11. Compose – Build & Scale

Command	Description
<code>docker compose build</code>	Build/rebuild services
<code>docker compose build --no-cache</code>	Build without cache
<code>docker compose up --build</code>	Rebuild then start
<code>docker compose pull</code>	Pull service images
<code>docker compose push</code>	Push service images
<code>docker compose up -d --scale <service>=3</code>	Scale a service
<code>-f <file.yml></code>	Use a specific compose file

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Kubernetes

`kubectl` is the primary CLI. Tip: `alias k=kubectl`.

12. Cluster & Context

Command	Description
<code>kubectl cluster-info</code>	Show cluster endpoints
<code>kubectl version</code>	Client/server version
<code>kubectl config get-contexts</code>	List contexts
<code>kubectl config current-context</code>	Show current context
<code>kubectl config use-context <name></code>	Switch context
<code>kubectl config set-context --current --namespace=<ns></code>	Set default namespace
<code>kubectl get nodes</code>	List cluster nodes
<code>kubectl get namespaces</code>	List namespaces

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13. Pods, Deployments & Workloads

Command	Description
<code>kubectl get pods</code>	List pods
<code>kubectl get pods -A</code>	All namespaces
<code>kubectl get pods -o wide</code>	Extra details (node, IP)
<code>kubectl get deployments</code>	List deployments
<code>kubectl get all</code>	List common resources
<code>kubectl run <name> --image=<image></code>	Run a pod
<code>kubectl create deployment <name> --image=<image></code>	Create a deployment
<code>kubectl delete pod <name></code>	Delete a pod
<code>kubectl get rs</code>	List replica sets
<code>kubectl get jobs/cronjobs</code>	List jobs/cronjobs

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14. Inspecting & Debugging (kubectl)

Command	Description
<code>kubectl describe pod <name></code>	Detailed pod info & events
<code>kubectl logs <pod></code>	View pod logs
<code>kubectl logs -f <pod></code>	Follow logs
<code>kubectl logs <pod> -c <container></code>	Specific container logs
<code>kubectl logs --previous <pod></code>	Logs from crashed instance
<code>kubectl exec -it <pod> -- sh</code>	Shell into a pod
<code>kubectl exec <pod> -- <cmd></code>	Run a command in a pod
<code>kubectl get events --sort-by=.lastTimestamp</code>	Recent events
<code>kubectl top pod/top node</code>	Resource usage (metrics)
<code>kubectl cp <pod>:/path ./local</code>	Copy out of a pod
<code>kubectl debug <pod> -it --image=busybox</code>	Ephemeral debug container

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15. Services & Networking

Command	Description
<code>kubectl get svc</code>	List services
<code>kubectl expose deployment <name> --port=80 --type=ClusterIP</code>	Create a service
<code>kubectl get ingress</code>	List ingresses
<code>kubectl get endpoints</code>	List endpoints
<code>kubectl port-forward <pod> 8080:80</code>	Forward a local port
<code>kubectl port-forward svc/<name> 8080:80</code>	Forward to a service

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16. Config, Secrets & Storage

Command	Description
<code>kubectl get configmaps</code>	List ConfigMaps
<code>kubectl create configmap <name> --from-file=<f></code>	ConfigMap from file
<code>kubectl create configmap <name> --from-literal=K=V</code>	ConfigMap from literal
<code>kubectl get secrets</code>	List secrets
<code>kubectl create secret generic <name> --from-literal=K=V</code>	Create a secret
<code>kubectl get pvc</code>	List persistent volume claims
<code>kubectl get pv</code>	List persistent volumes
<code>kubectl get sc</code>	List storage classes

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17. Scaling, Rollouts & Updates

Command	Description
<code>kubectl scale deployment <name> --replicas=3</code>	Scale a deployment
<code>kubectl autoscale deployment <name> --min=2 --max=10 --cpu-percent=80</code>	Autoscale (HPA)
<code>kubectl set image deployment/<name> <c>=<image>:<tag></code>	Update image
<code>kubectl rollout status deployment/<name></code>	Watch a rollout
<code>kubectl rollout history deployment/<name></code>	Rollout history

Command	Description
<code>kubectl rollout undo deployment/<name></code>	Roll back
<code>kubectl rollout restart deployment/<name></code>	Restart pods
<code>kubectl edit deployment <name></code>	Edit live resource

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18. Apply, Delete & YAML

Command	Description
<code>kubectl apply -f <file.yaml></code>	Create/update from file
<code>kubectl apply -f <dir>/</code>	Apply a directory
<code>kubectl apply -k <dir>/</code>	Apply with Kustomize
<code>kubectl delete -f <file.yaml></code>	Delete from file
<code>kubectl delete <type> <name></code>	Delete a resource
<code>kubectl get <type> <name> -o yaml</code>	Export as YAML
<code>kubectl create <type> <name> --dry-run=client -o yaml</code>	Generate YAML template
<code>kubectl explain <resource></code>	Field documentation
<code>kubectl diff -f <file.yaml></code>	Preview changes

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